

Patient Photo

REF. DOCTOR : doctor@example.com

PATIENT : rajen | ID: #228540

AGE/SEX : 45 Years

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Test Report Status	Final	Results	Reference Interval	Units
AI-BASED TUBERCULOSIS DIAGNOSTIC ANALYSIS (TBFERON-AI)				
SPECIMEN SOURCE		MULTIMODAL INPUT		
METHOD : DIGITAL CHEST X-RAY + CLINICAL DATA				
CNN X-RAY PROBABILITY		63.7%	< 40.0%	%
METHOD : DEEP LEARNING - MOBILENETV2				
CLINICAL PARAMETER SCORE		99.0%	< 40.0%	%
METHOD : LOGISTIC REGRESSION ML MODEL				
FINAL FUSED TB PROBABILITY		77.8% High	< 40.0%	%
METHOD : WEIGHTED MULTIMODAL FUSION (60% IMAGE + 40% CLINICAL)				
COUGH DURATION		0 Weeks	> 2 Weeks indicates risk	Weeks
METHOD : CLINICAL PARAMETER				
FEVER / WEIGHT LOSS / NIGHT SWEATS		Yes / Yes / Yes		
METHOD : CLINICAL PARAMETER				
SPUTUM SMEAR MICROSCOPY		Positive	NEGATIVE	
METHOD : LAB DATA / ACID-FAST BACILLI				
GENEXPERT MTB/RIF		Positive	NEGATIVE	
METHOD : LAB DATA / NUCLEIC ACID AMPLIFICATION				
INTERPRETATION		POSITIVE	NEGATIVE	
METHOD : AI MULTIMODAL FUSION ANALYSIS				

LUNG REGION CLASSIFICATION (MedSAM AI Segmentation)		
Anatomical Region	Severity	Status
Upper Right Lobe	HIGH	Critically Affected
Upper Left Lobe	MEDIUM	Moderately Affected
Mid Right Lobe	MEDIUM	Moderately Affected
Mid Left Lobe	LOW	Mildly Affected
Lower Lobes	HIGH	Critically Affected

Interpretation(s)

The AI-based TB detection system uses a multimodal approach combining Deep Learning analysis of the Chest X-Ray image with clinical parameter scoring (cough duration, fever, weight loss, night sweats) to arrive at a fused probability score.

AI Recommendation: Immediate medical evaluation required. Isolate patient and schedule sputum test.

DISCLAIMER: This report is generated by an AI system for clinical decision support only. It is not intended to replace professional medical diagnosis. A qualified physician should evaluate these results in the context of the patient's full clinical picture. Confirmatory tests (sputum culture, GeneXpert, TST) are recommended for definitive TB diagnosis.

Authorized Signatory

TB-Vision AI Diagnostic System

Reviewing Physician

doctor@example.com



